

JUDE WATIMONGO

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PROFILE SUMMARY

A highly driven Master's in Mechatronics Engineering student with a strong foundation in hardware development and embedded systems, particularly in innovative prototype design. Proficient in energy management and renewable energy, with hands-on expertise in hybrid vehicles and solar technology. An active IEEE member with strong coding skills and problem-solving capabilities, ready to leverage technical knowledge and practical experience in impactful projects.

EDUCATION

Ashesi University

MPhil. Mechatronics Engineering

Cumulative GPA: 3.67/4.00

Relevant Courses: Control Systems, Robotics, Machine Learning, Energy Systems and Mobility, Industrial Automation

Berekuso, Eastern Region

Expected Date of Graduation: June 2027

Ashesi University

BSc. Electrical and Electronics

Cumulative GPA: 3.34/4.00

Relevant Courses: System Dynamics, Leadership, Programming, Circuit Design, Fundamentals of Design and Entrepreneurship

Berekuso, Eastern Region

Date of Graduation: June 2023

WORK EXPERIENCE

Sunbird AI, Uganda

June 2022 – Aug 2022

Hardware intern

- Developed prototype noise monitoring sensors utilising microcontrollers for effective environmental assessment.
- Implemented a voice feedback recording device prototype for an NGO called SEMA, enhancing public service feedback.
- Conducted rigorous testing of embedded systems to validate performance against established metrics in measuring sound pollution and calibrating the sensors.
- Presented a comprehensive account of the final solution to the board, resulting in approval and further project development.

PROJECT AND RESEARCH

Hybrid Electric Vehicle Energy Management Project

June - July 2025

Energy Optimisation project

- Designed and implemented an optimal EMS for a P2 hybrid vehicle using Pontryagin's Minimum Principle and ECMS, enabling real-time optimisation of engine-motor power split to reduce fuel consumption and maintain battery SOC.
- Built a MATLAB/Simulink controller that improved SOC tracking through a PI-based adaptive equivalence factor, achieving accurate SOC control on WLTC and SORT cycles and demonstrating strong controller robustness.
- Delivered measurable performance outcomes, including fuel-efficiency improvements across driving cycles and validated EMS behaviour through multi-cycle simulation analysis.

Solar Drier design

Sept 2021 – Nov 2021

Class Team Project

- Designed a solar drier using SolidWorks and later built it in the workshop using plywood and glass
- Researched about solar energy and made appropriate calculations for the design of a product to be used by a community in Agbokpa

Kwadkopter Team.

Sept 2019 – Nov 2019

Team member

- Developed Arduino code for drone functionality.
- Created and fabricated a lightweight aluminium frame using SolidWorks and thereafter in the workshop.
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CO-CURRICULAR ACTIVITIES

IEEE Professional Organisation, Ghana Section

Feb 2025 – present

Graduate Student Member

- Competed in the IEEE Code Xtreme competition, achieving a top 8 ranking among teams in Ghana, demonstrating strong coding and problem-solving skills.
- Engaged with the IEEE Robotics and Automation Society, enhancing my expertise and contributing to knowledge-sharing initiatives within the field.

Resident Coordinator, Ashesi University

Dec 2025 – Jan 2026

- Served as the first point of call to 150 students and the hostel's management to meet all logistical needs on campus, gaining a rating of 3.82/5.
- Liaised with service providers to resolve conflicts of interest with student interests within 2 days, improving living conditions on campus.

TECHNICAL SKILLS

- Programming Languages: Python, C/C++, MATLAB, Java
- Software/Tools: SolidWorks, Simulink, Arduino, PLC Programming, Proteus circuit design

REFERENCES

Available upon request